Mock Paper A(2)

Started on	Monday, 28 May 2018, 2:55 AM
State	Finished
Completed on	Monday, 11 June 2018, 5:30 AM
Time taken	14 days 2 hours
Overdue	14 days 1 hour
Grade	**0.00** out of 100.00
Feedback	This is a bit worrying. You need to prioritize your revision urgently.

Question 1
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 001
What percentage of patients on clozapine develop agranulocytosis during the first year of treatment?

Select one:
- 0.15
- 0.1
- 0.05
- 0.01
- 0.2

Your answer is incorrect.

In an epidemiologic drug registry study in the United Kingdom, it was found that clozapine causes neutropenia in 1.5% to 2.9% of patients treated over a 1-year period, with progression to agranulocytosis in 0.8% of patients

The correct answer is: 0.01

Question 2
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 002
The annual incidence of tardive dyskinesia in patients with chronic schizophrenia who are on any antipsychotic treatment is

Select one:
- 0.05
- 0.03
- 0.1
- 0.01
- 0.2

Your answer is incorrect.

Longitudinal incidence data with first generation antipsychotics (FGAs) as a class suggest that the annual incidence of tardive dyskinesia risk is ~4% per year of exposure. Thus, of 100 patients started on haloperidol or chlorpromazine, four would have some manifestations of tardive dyskinesia at the end of 1 year. That rate increases after the second or third year. It is unknown for how long this rate remains linear and if it would extend to 100% (Retrieved from http://primarypsychiatry.com/in-session-with-jeffrey-a-lieberman-md-tardive-dyskinesia/). A systematic review published in 2004 (Corell et al., 2004) suggested that weighted mean annual incidence of tardive dyskinesia for second-generation antipsychotics was 0.8% (range=0.0%-1.5%) in adults and 5.3% in patients aged 54 years and older, compared to 5.4% in adults treated with haloperidol. A large prospective study of 9298 patients (SOHO study) suggested that overall, the risk of TD in patients taking any antipsychotic is likely to be around 3% (Tenback et al., 2006).

The correct answer is: 0.03

Question 3
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 004
The risk of fetal malformations with sodium valproate is approximately

Select one:
- 7%
- 28%
- 1%
- 70%
- 90%

Your answer is incorrect.

Epilepsy itself increases the risk of foetal malformations two to three times when compared to the general population. 7.2% of pregnant mothers using valproate will have children showing some form of fetal malformation. In particular, the incidence of neural tube defects is 1% to 2%. This risk increases with increasing doses of valproic acid especially when the dose is more than 1gm/day.

NICE. Technology Appraisal 76.

The correct answer is: 7%

Question 4
Not answered
Marked out of 1.00

MoK Adverse Effects 005
Which of the following drugs is currently thought to cause least QT prolongation of all antipsychotics?

Select one:

Flag question

- Olanzapine
- Quetiapine
- Haloperidol
- Clozapine
- Aripiprazole

Your answer is incorrect.

Current available evidence favours aripiprazole though this may be revised in future with more data becoming available.

The correct answer is: Aripiprazole

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 009

According to the CATIE study of patients with schizophrenia, the number of patients meeting criteria for metabolic syndrome ranges between what percentages?

Select one:

- 10-20%
- 31-40%
- 51-60%
- 21-30%
- 41-50%

Your answer is incorrect.

In the CATIE study, the metabolic syndrome prevalence was found to be 51.6% for females and 36.0% for males with schizophrenia.

The correct answer is: 41-50%

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 010

The upper normal QTc level in female subjects is

Select one:

- 440 ms
- 450 ms
- 500 ms
- 470 ms
- 420 ms

Your answer is incorrect.

Normal QTc for men is 440 ms and 470 ms for women. There is a clearly increased risk of arrhythmias in QTc values over 500 ms.

The correct answer is: 470 ms

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 011

A 44-year-old woman on clozapine has gained 15Kgs in 3 months after starting the treatment. This antipsychotic induced weight gain is most likely to be due to;

Select one:

- Hypoprolactinaemia
- Leptin sensitisation
- 5-HT2C antagonism
- M4 blockade
- H2 antagonism

Your answer is incorrect.

Agonists at the 5HT2 family of receptors cause hypophagia and can cause weight loss in animals and human beings. In rats, 5HT2C antagonists cause an increase in feeding. Clozapine is a potent 5HT2C antagonist. Autoradiographic studies have shown that 5HT2C receptors are localised in high density in the ventromedial and dorsomedial nuclei within the hypothalamic satiety control centres. 5HT2C knockout mice are overweight and exhibit increased feeding as opposed to metabolic changes.3 Clozapine may disrupt serotonergic pathways, resulting in weight gain in predisposed people.

The correct answer is: 5-HT2C antagonism

Question 8

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 013

Which of the following is true with regard to acetylcholine (Ach) ?

Select one:

- Acetylcholinergic projections are more predominant than serotonergic projections in the cortex
- Tricyclics increase acetylcholine levels
- Antidementia drugs reduce acetylcholine concentration
- Acetylcholine agonists are useful in mild to moderate dementia of Alzheimer's type
- Acetylcholine is synthesised by acetyl cholinesterase

Your answer is incorrect.

The distribution of cholinergic innervation in the human cerebral cortex reveals that all cytoarchitectonic regions and layers of the cerebral cortex display a dense cholinergic innervation The density of cholinergic axons is higher in the more superficial layers (layers I and II and the upper parts of layer III) of the cerebral cortex. Distinct patterns of lamination exist in individual cytoarchitectonic regions (Structure and Function of Cholinergic Pathways. http://www.acnp.org/g4/GN401000012/CH012.html (accessed April 13, 2015)).

The correct answer is: Acetylcholinergic projections are more predominant than serotonergic projections in the cortex

Question 9

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 014

Which of the following antidepressants is least likely to be associated with sexual dysfunction in male depressed patients?

Select one:

- Agomelatine
- Trazodone
- Venlafaxine
- Fluoxetine
- Amitriptyline

Your answer is incorrect.

The overall side-effect profile of agomelatine appears relatively mild. For example, agomelatine has no discontinuation syndrome and is generally weight neutral. Agomelatine appears to be relatively free of sexual dysfunction and related side effects. (Sansone & Sansone, 2011)

The correct answer is: Agomelatine

Question 10

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 017

A 66 year old man taking antidepressant therapy develops dizziness every morning. He gives a history of palpitations accompanied by the dizziness. There is no history of vertigo or falls but the dizziness improves spontaneously on reclining. Which of the following receptor blockade can explain the above symptoms?

Select one:

- Alpha 1 adrenergic
- Cholinergic nicotinic
- Cholinergic muscarinic
- Alpha 2 adrenergic
- Beta adrenergic

Your answer is incorrect.

Alpha 1 blockade can cause dizziness, postural hypotension, ejaculatory disturbance and dependent edema.

The correct answer is: Alpha 1 adrenergic

Question 11

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 018

Mr. X is taking trazodone for his depressive symptoms. He complains of excessive sedation. This effect is most likely to be due to the blockade of which of the following receptors?

Select one:

- Nicotinic
- Beta adrenergic
- Alpha adrenergic
- Muscarinic
- Histaminic

Your answer is incorrect.

Trazodone's potent alpha1-adrenergic blockade (~3-fold lower relative to 5-HT2A) may cause some side effects like sedation and orthostatic hypotension. Conversely, along with 5-HT2A antagonism, it may underlie trazodone's efficacy as a hypnotic. This seems possible as trazodone's antihistamine activity is relatively weak and probably clinically insignificant; hence, it cannot explain trazodone's sleep-inducing/enhancing effects.

The correct answer is: Alpha adrenergic

Question 12

Not answered

Marked out of 1.00

Flag question

MoK Basic Pharmacology 001

Choose the most accurate statement about cannabis

Select one:

- Cannabis commonly causes physical dependence.
- 9-delta-tetrahydrocannabinol is the active substance
- Fatal overdose is reported in several countries
- Urine tests usually remain positive for up to a week after regular use has ceased
- Anandamide is a specific cannabinoid receptor and a naturally occurring antagonist at this receptor

Your answer is incorrect.

There are 60 psychoactive cannaboids, the most important of which is 9-delta-tetrahydrocannabinol (THC). Anandamide is a specific cannabinoid receptor and a naturally occurring agonist at this receptor. Cannabis is not usually associated with physical dependence. Urine tests usually remain positive for up to 4weeks after regular use has ceased. Unlike most other drugs of abuse, there are no reports of fatal overdose from cannabis.

The correct answer is: 9-delta-tetrahydrocannabinol is the active substance

Question 13
Not answered
Marked out of 1.00
Flag question

MoK Basic Pharmacology 002
Mr.Z is a 26 year old gentleman who was diagnosed with paranoid schizophrenia 4 years ago and has had 6 hospitalisations over last 3 year period. Most of his relapses have been associated with non adherence to medication. Which of the following interventions are most likely to improve compliance for this patient?

Select one:
- Insight orientated psychotherapy
- Cognitive motivational intervention
- Supportive psychotherapy
- Family therapy
- Psycho educational intervention

Your answer is incorrect.

Studies on psycho social intervention for improving medication adherence showed that psycho education alone and family therapy alone were ineffective and programmes that focus on the attitudinal and behavioural aspects of taking medications have better outcomes. There is no evidence based data available for supportive psychotherapy and insight orientated psychotherapy for improving medication adherence.

The correct answer is: Cognitive motivational intervention

Question 14
Not answered
Marked out of 1.00
Flag question

MoK Basic Pharmacology 003
Carlsson is associated with which of the following?

Select one:
- learned helplessness hypothesis of depression
- hairline triggers in borderline personality
- endocrine hypothesis of depression
- kindling hypothesis of depression
- dopamine hypothesis of schizophrenia

Your answer is incorrect.

Arvid Carlsson made the seminal discovery in the late 1950's that dopamine is a transmitter in the mammalian brain.

The correct answer is: dopamine hypothesis of schizophrenia

Question 15
Not answered
Marked out of 1.00
Flag question

MoK Basic Pharmacology 004
The risk factor predicting poor response to lithium prophylaxis in bipolar illness is

Select one:
- Dominant personality
- Social support
- High social status
- High age of illness onset
- High number of previous hospitalizations

Your answer is incorrect.

Predictors of poor response to lithium in bipolar prophylaxis include High number of previous hospitalization, stress, high expressed emotion, and high number of life events, episodic patterns of depression- mania interval, continuous cycling, and unemployment and neurotic personality.

The correct answer is: High number of previous hospitalizations

Question 16
Not answered
Marked out of 1.00
Flag question

MoK Basic Pharmacology 005
Which of the following antipsychotics is recommended for mixed affective states in bipolar disorder?

Select one:
- perphenazine
- amisulpride
- sulpride
- quetiapine
- Olanzapine

Your answer is incorrect.

Though based on a limited amount of information, a Cochrane review (and British Association of Psychopharmacology guidelines) suggests that there is evidence that olanzapine may prevent further mood episodes in patients who have responded to olanzapine during an index manic or mixed episode and who have not previously had a satisfactory response to lithium or valproate (http://www.ncbi.nlm.nih.gov/pubmedhealth/PMH0012734/)

The correct answer is: Olanzapine

Question 17
Not answered

MoK Basic Pharmacology 007
A 36-year-old cocaine user notices that the longer he uses the drug, the more of it he requires to achieve the same effect. This is best described as

Marked out of 1.00

Flag question

Select one:

- Addiction
- Dependence
- Tolerance
- Sensitisation
- Compliance

Your answer is incorrect.

Tolerance refers to the pharmacological adaptation because of which a larger dose of a drug becomes necessary over time to achieve the same effect.

The correct answer is: Tolerance

Question 18

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 001

Which of the following is true with regard to nicotinic receptors?

Select one:

- Blockade causes problems with fine motor movements
- Mediate anti-parkinsonian effects of amantadine
- All nicotinic receptors are excitatory
- Mediate sympathomimetic actions of dopamine
- Anticholinergic effects of antipsychotics are mediated by nicotinic receptors

Your answer is incorrect.

All nAChR are excitatory. Nicotinic acetylcholine receptors (nAChRs) are excitatory neurotransmitter-gated ion channels and members of a superfamily that also includes ionotropic receptors such as GABA-A. The other type of cholinergic receptors, muscarinic group, mediate anti-cholinergic effects of psychotropics and disturbances of fine motor movements seen in certain neurological disorders. Nicotinic receptors when blocked produce gross neuromuscular paralysis.

The correct answer is: All nicotinic receptors are excitatory

Question 19

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 002

SSRIs, on stimulating which receptors, will induce sleep disturbances?

Select one:

- 5-HT2A
- 5-HT3
- 5-HT1A
- H-1
- 5-HT1D

Your answer is incorrect.

5-HT2A stimulation by SSRIs will lead to insomnia

The correct answer is: 5-HT2A

Question 20

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 003

Aripiprazole has an antagonistic effect on which of the following receptors?

Select one:

- D2 receptors
- 5-HT3 receptors
- 5-HT2A receptors
- 5-HT1A receptors
- D1 receptors

Your answer is incorrect.

Aripiprazole is a D2 partial agonist, an action that is unique to aripiprazole; it is also a 5-HT2A antagonist and 5-HT1A partial agonist. Minor actions include H1 antagonism, alpha-1, alpha-2 antagonism and to some extent diffuse muscarininc antagonism. (M1, M2, M3, M4, M5 antagonist)

The correct answer is: 5-HT2A receptors

Question 21

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 004

Agonistic effect of fluoxetine on which of the following receptors results in anorgasmia?

Select one:

- Alpha-1 adrenergic receptors
- 5HT-1A stimulation
- D-2 receptors
- 5HT-2C stimulation
- Mu opiod receptors

Your answer is incorrect.

5-HT2A/2C stimulation by SSRIs can cause delayed ejaculation and Anorgasmia.

The correct answer is: 5HT-2C stimulation

Question 22

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 005
Which of the following is a noncompetitive NMDA blocker?

Select one:

- Lamotrigine
- Bupropion
- Acamprosate
- Disulfiram
- Varenicline

Your answer is incorrect.

Acamprosate has a structure similar to that of GABA and is active at the N-methyl-D-aspartate (NMDA) receptor (Knopfel et al 1987), via non-competitive NMDA blockade. But this action is supposed to be weak in therapeutic doses.

The correct answer is: Acamprosate

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 001
The number of half-lives normally taken by drugs to reach steady plasma levels is;

Select one:

- 7
- 9
- 2
- 3
- 5

Your answer is incorrect.

In clinical practice, after regular administration, a drug takes 4 to 5 times the half-life for its serum concentration to reach steady state.

The correct answer is: 5

Question 24

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 002
Which of the following is correct with regard to benzodiazepine metabolism?

Select one:

- Desmethyldiazepam is an inactive metabolit
- Diazepam achieves nearly 10 times greater plasma level in the elderly
- Diazepam is around 95% protein bound
- Duration of action depends primarily on elimination half-life
- Elimination half-life is reduced in the elderly

Your answer is incorrect.

Diazepam produces an active metabolite desmethyldiazepam (also called as nordiazepam - t1/2 71 hours) via CYP3A4 and 2C19. Diazepam is heavily protein bound (around 95%). This plasma protein binding is significantly reduced in the elderly subjects, but the plasma free (unbound) diazepam concentrations do not significantly exceed when compared to young adults. The elimination half-life is increased by 100% in elderly (around 90 hrs) as clearance is reduced, leading to a much prolonged duration of action than in younger adults (around 46 hrs). Duration of action of benzodiazepines depends on rate and extent of distribution, in addition to half-life. Most long-acting agents have long elimination half-life.

The correct answer is: Diazepam is around 95% protein bound

Question 25

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 004
Mr Y is a 27 year old gentleman who has recently been started on lithium for treatment of his acute phase of his manic episode. What is the therapeutic blood lithium level you would aim for?

Select one:

- 0.6- 0.8 M Eq/ L
- 0.4- 0.5 M Eq/ L
- 0.2- 0.4 M Eq/ L
- 1.5 -2.0 M Eq/ L
- 0.8-1.2 M Eq/ L

Your answer is incorrect.

Lithium is currently the drug of choice for treatment of the acute phase of a manic episode as well as for prevention of further episodes of bipolar mood disorder. The treatment is closely monitored by repeated blood levels as the difference between therapeutic and lethal blood levels is not very wide. Therapeutic blood levels for antimanic or antidepressant effect range between 0.8 and 1.2 MEq/L; Prophylactic blood lithium levels vary from 0.6 to 1.2 MEq/L. A blood lithium level of more than 2 milliequivalents per litre may be associated with toxicity while a level of more than 2.5 may be lethal.

The correct answer is: 0.8-1.2 M Eq/ L

Question 26
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 005
The minimum effective dose of Aripiprazole in treating a psychotic episode is

Select one:
- 20mg
- 15mg
- 10mg
- 25mg
- 5mg

Your answer is incorrect.

Aripiprazole was shown to be superior to placebo at doses of 15 mg/day, 20 mg/day, and 30 mg/day in at least two clinical trials for each dose and at 10 mg/day in another trial. Two studies that evaluated the efficacy of aripiprazole at lower doses (2 and 5mg/day) in schizophrenia, failed to demonstrate the superiority of aripiprazole versus placebo. The available trial evidences indicate that Aripiprazole 10 mg/day is the minimum effective dose for the treatment of patients with schizophrenia experiencing acute relapse. Aripiprazole 5 mg/day produced improvements in some outcome measures but not in the primary efficacy measure, suggesting that it may lack efficacy for treating acute psychosis. CNS Spectr. 2006;11(9):691-702

The correct answer is: 10mg

Question 27
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 006
The minimum duration of gap recommended between stopping venlafaxine and starting duloxetine is

Select one:
- 2 weeks
- 2 months
- 2 days
- 1 day
- 1 week

Your answer is incorrect.

According to product info, a 2 weeks gap is recommended though popular prescribing guidelines suggest 2 weeks of tapering for venlafaxine till stopping it, followed by the alternate day prescription of duloxetine.

The correct answer is: 2 weeks

Question 28
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 013
The rate-limiting enzyme in the metabolism of alcohol is;

Select one:
- Alcohol dehydrogenase
- Acetyl co A dehydrgenase
- Aldehyde dehydrogenase
- Acetic acid
- Acetyl dehydrogenase

Your answer is incorrect.

The major site of alcohol metabolism is the liver. Alcohol dehydrogenase is the rate-limiting enzyme in the metabolism of alcohol which makes it impossible to speed up alcohol metabolism.

The correct answer is: Alcohol dehydrogenase

Question 29
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 014
Which of the following drugs is least dependent on hepatic metabolism?

Select one:
- Chlorpromazine
- Clozapine
- Olanzapine
- Amisulpride
- Risperidone

Your answer is incorrect.

Renal elimination without significant liver breakdown is seen with amisulpride and sulpride (90% elimination via renal route)

The correct answer is: Amisulpride

Question 30
Not answered

MoK Pharmacokinetics 016
Which of the following drugs have a half-life of 3 days?

Marked out of 1.00

Flag question

Select one:

- Carbamazepine
- Aripiprazole
- Quetipaine
- Reboxetine
- Fluoxetine

Your answer is incorrect.

Aripiprazole displays linear kinetics and has a mean terminal half-life of 60 hours. (Range 46-68 hours)

The correct answer is: Aripiprazole

Question 31

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 001
Which of the following is a localising sign for frontal lesions?

Select one:

- Postural headache
- Ptosis
- Nystagmus
- Olfactory disturbances
- Myoclonus

Your answer is incorrect.

The most classic presentation of frontal lobe dysfunction is an olfactory groove meningioma characterized by anosmia, loss of inhibition, memory impairment, headaches, and visual symptoms. But anosmia itself is not a good localising sign as often the cause of olfactory disturbance is peripheral - related to nostrils (Frontal Lobe Syndromes Clinical Presentation, http://emedicine.medscape.com/article/1135866-clinical (accessed April 13, 2015).).

The correct answer is: Olfactory disturbances

Question 32

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 002
National adult reading test is used to assess

Select one:

- Visuospatial ability
- Premorbid intelligence
- Attentional bias
- Nonverbal intelligence
- Frontal lobe functions

Your answer is incorrect.

The validity of NART as a measure of premorbid ability rests upon the assumptions that reading ability (of irregular words) is relatively independent of brain damage, and that it is a strong predictor of intelligence in the normal population.The test requires subjects to read out loud a set of 50 words which are irregular in terms of their grapheme-phoneme correspondences (Coltheart et al., 1987). The responses are individually scored as correct or incorrect, according to their pronunciation. This score can then be used to derive a premorbid IQ estimate. Journal of the International Neuropsychological Society (2002), 8, 847-854.

The correct answer is: Premorbid intelligence

Question 33

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 003
A 34-year-old man unable to remember any of the events that happened at the hospital during the time of his admission for a head injury even after two years after recovery. He has no other memory problems. This is

Select one:

- Confabulation
- Anterograde amnesia
- Retrograde amnesia
- Episodic memory loss
- Dissociative memory loss

Your answer is incorrect.

After the head injury, he had a period of anterograde amnesia during which he did not form any memory trace for a period of time. As a result, he has no memory of these events even after two years.

The correct answer is: Anterograde amnesia

Question 34

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 004
A 70-year-old man presents with sudden onset of right arm weakness following severe headache and inability to understand speech. His speech problem is accurately described as

Select one:

- Aphasia
- Alexia

- Aphemia
- Agraphia
- Amusia

Your answer is incorrect.

This is receptive aphasia (Wernicke's area damage)

The correct answer is: Aphasia

Question 35

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 005

On routine examination of eyes, accommodation reflexes were present but pupils were small irregular with absent light reflexes. This type of pupil is called

Select one:

- Marcus Gunn pupil
- Senile pupil
- Anosognosia
- Kayser-Fleischer pupil
- Argyll Robertson pupils

Your answer is incorrect.

Douglas Argyll Robertson's (1837-1909) conducted experimental work with physostigmine in 1863. He observed patients with spinal cord disease who had lost the response to light even though accommodation to near objects was normal. This phenomenon, known subsequently as the Argyll Robertson pupil came to be considered pathognomonic of tabes dorsalis, general paresis and neurovascular syphilis.

The correct answer is: Argyll Robertson pupils

Question 36

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 006

Abnormalities in Stroop test suggests

Select one:

- Short term memory disturbance
- Damage to occipital cortex
- Attentional bias
- Cortical blindness
- Lesion of optic chiasma

Your answer is incorrect.

An abnormal Stroop effect with the loss of appropriate inhibitory control is seen in frontal lobe lesions. Abnormalities in modified emotional Stroop interference effects (e.g. on threat words in anxious subjects on the emotional Stroop task) have generally been interpreted as evidence for mood-congruent attentional bias in various mental states.

The correct answer is: Attentional bias

Question 37

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 007

Which of the following instruments can be used to screen for cognitive deficits after head injury?

Select one:

- Bender visual motor gestalt test
- National adult reading test
- Glasgow coma scale
- Weschler adult intelligence scale
- Raven's progressive matrices

Your answer is incorrect.

The Bender-Gestalt is used to evaluate the maturity of the visual-motor system and to screen children for developmental delays. The test is also used to assess brain damage and neurological deficits. Individuals who have suffered a traumatic brain injury may be given the Bender-Gestalt as part of a battery of neuropsychological measures or tests.

The correct answer is: Bender visual motor gestalt test

Question 38

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 008

A 64-year-old man has behavioural changes with irritability. He is found wandering in the streets near his home. On neurological examination, there is evidence of mild aphasia and cognitive dysfunction, but motor function is preserved. He seems to have no awareness regarding his difficulties and has a silly, jocular affect. Which of the following is most likely to be seen in CT/MRI scan?

Select one:

- Dilated ventricles without cerebral atrophy
- Predominant occipito parietal atrophy
- Diffuse cortical atrophy with apparently dilated ventricles
- Bilateral frontal atrophy
- Space occupying lesion obstructing third ventricle

Your answer is incorrect.

The presence of reduced verbal fluency, loss of insight and having a silly affect points towards frontal lesions.

The correct answer is: Bilateral frontal atrophy

Question 39

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 009
Nonfluent aphasia is associated with lesions in or near which of the following brain regions?

Select one:

- Parietal operculum
- Anterior cingulate
- Superior temporal gyrus
- Insula
- Broca's area

Your answer is incorrect.

Nonfluent aphasia is associated with lesions in or near Broca's area in motor cortex. Temporal and parietal lobes are usually not impaired. Therefore, receptive language and comprehension are not usually impaired in nonfluent aphasia.

The correct answer is: Broca's area

Question 40

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 001
What is the concordance rate of schizophrenia in monozygotic twins?

Select one:

- 0.22
- 0.48
- 0.18
- 0.14
- 0.08

Your answer is incorrect.

Most studies suggest that the concordance rates are 44-50% for MZ twins and 10-15% for DZ twins. On the average, most studies show that the concordance rate for schizophrenia is three times higher in monozygotic twins than in dizygotic twins.

The correct answer is: 0.48

Question 41

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 002
The gene implicated in Huntington's disease is present on the chromosome

Select one:

- 21
- 14
- 7
- 4
- 1

Your answer is incorrect.

Huntington's disease-related gene is located on the short arm of chromosome 4; it contains a variable stretch of repeating CAG trinucleotides which code for a glutamine residue in the huntingtin protein. Individuals with 40 or more repeats will develop Huntington's disease; those with 35 or fewer will not. But how this polyglutamine expansion in huntingtin causes the disease is uncertain. Am J Psychiatry 164:1318, September 2007

The correct answer is: 4

Question 42

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 003
Which of the following is associated with narcolepsy?

Select one:

- Waxy flexibility
- Unrefreshing sleep attacks
- 2% prevalence
- Prolonged REM latency
- HLA-DR2

Your answer is incorrect.

Narcolepsy is associated with cataplexy (NOT catalepsy / waxy flexibility - the latter are catatonic features). It is a disabling sleep disorder affecting 0.02% of adults worldwide. It is characterised by severe, irresistible daytime sleepiness and sudden loss of muscle tone (cataplexy), and can be associated with sleep-onset or sleep-offset paralysis and hallucinations, frequent movement and awakening during sleep, and weight gain. Sleep monitoring during night and day shows rapid sleep onset and abnormal, shortened rapid-eye-movement sleep latencies. The onset of narcolepsy with cataplexy is usually during teenage and young adulthood and persists throughout the lifetime. Pathophysiological studies have shown that the disease is caused by the early loss of neurons in the hypothalamus that produce hypocretin, a wakefulness-associated neurotransmitter present in the cerebrospinal fluid. The cause of neural loss could be autoimmune since most patients (more than 90%) have the HLA DQB1*0602 allele that

predisposes individuals to the disorder. The frequency of HLA-DR2 is also increased in narcolepsy though not as high as HLA DQB1*0602. Treatment is with stimulant drugs to suppress daytime sleepiness, antidepressants for cataplexy, and gamma-hydroxybutyrate for both symptoms (Narcolepsy with cataplexy, http://www.ncbi.nlm.nih.gov/pubmed/17292770).

The correct answer is: HLA-DR2

Question 43
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 005
Mr. Y has a well-established diagnosis of bipolar affective disorder. His younger brother, who is also his main carer, is aware that this disorder has a strong genetic loading and is worried. What is the prevalence rate in first-degree relatives of patients with bipolar affective disorder?

Select one:
- 15-20%
- 0.4
- 1-2%
- 0.05
- 0.79

Your answer is incorrect.

Bipolar disorder has a multifactorial aetiology. Both family studies and twin studies have demonstrated a strong genetic loading for it. A prevalence rate of 15-20% has been demonstrated in first-degree relatives of patients with the bipolar affective disorder. Twin studies have shown a concordance rate of up to 79% in Monozygotic twins and 19% in dizygotic twins. Ref: Pandarakalam, JP (2008) Diagnostic conundrums of bipolar disorder. Progress in Neurology and Psychiatry, 12, 6-11.

The correct answer is: 15-20%

Question 44
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 006
Mr. Y is a 40-year-old gentleman with history of callous unemotional traits. What is the heritability (genetic contribution) of these traits?

Select one:
- 5-10%
- 50-60%
- 1-2%
- 10-20%
- 30-40%

Your answer is incorrect.

Results from a meta-analysis of behaviour genetic studies on children with callous-unemotional traits and antisocial behaviour estimates, that 41% of the variance is due to genetic factors, 16% to shared environmental factors and 43% to non-shared environmental factors. Data from other studies appear to indicate moderate heritability for violent behaviour as well. (Ref: Viding E. On the Nature and Nurture of Antisocial Behavior and Violence. Retrieved from http://www.socialbehavior.uzh.ch/teaching/semsocialneurosciencews07/)

The correct answer is: 30-40%

Question 45
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 010
Which of the following substances is found in reduced levels in patients with high impulsive suicidality?

Select one:
- CYP4503A4
- COMT
- 5-HIAA
- BDNF
- Neuregulin

Your answer is incorrect.

Decreased 5-HIAA levels are associated with suicidal, antisocial and violent behaviours. [Halaban et al. 1996]. The psychological mechanism underlying this relationship is unclear, but some current models speculate that reductions in serotonin increase impulsivity and disinhibit behavior, thereby increasing the likelihood of engaging in risky and antisocial acts.

The correct answer is: 5-HIAA

Question 46
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 012
Which of the following types of chromosomes undergo Robertsonian translocation?

Select one:
- submetacentric
- Metacentric
- telocentric
- Holocentric
- acrocentric

Your answer is incorrect.

Robertsonian translocation is a non-reciprocal (i.e. unequal exchange) that results in a single fused chromosome from 2 acrocentric (non-homologous) chromosomes. A metacentric chromosome has centromere right in the middle. So p and q arms are of equal length. Most chromosomes are submetacentric where the arms are of unequal length. If the centromere is at the tail of a chromosome, it is called telocentric. With holocentric chromosomes, the entire length of the chromosome acts as the centromere.

These latter two types are not seen in humans.

The correct answer is: acrocentric

Question 47
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 013
Which of the following statements is true about sex linked recessive disorders?

Select one:
- 25% of sons of heterozygous mothers are affected
- 25% of daughters of heterozygous mothers are affected
- Females more affected than males
- All daughters of affected fathers are carriers of the disease
- Affected fathers pass disease to sons

Your answer is incorrect.

In sex-linked recessive disorders, an affected male can transmit the disease-causing mutation to his daughters, who remains normal phenotypically but carries and transmits the disease-causing allele to her sons. As females have two copies of the X chromosome, they need a double identical mutation for disease expression that is extremely rare.

The correct answer is: All daughters of affected fathers are carriers of the disease

Question 48
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 014
In Down's syndrome with Alzheimer's disease, the gene located is on the

Select one:
- Amyloid precursor protein
- Presenilin 1
- Presenilin 2
- Apolipoprotein e3
- Apolipoprotien e 4

Your answer is incorrect.

Individuals with Down syndrome develop beta-amyloid deposition characteristics of early onset Alzheimer's disease in mid-life, presumably due to an extra copy of the chromosome 21-located APP gene. By the age of 60, between 50 and 70% of people with Down syndrome develop dementia. Triplication of a short segment of APP in people without Down syndrome has also been associated with early-onset Alzheimer's disease (Retrieved from http://www.cdadc.com/ds/dementia.htm).

The correct answer is: Amyloid precursor protein

Question 49
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 017
In which of the following types of inheritance, 75% of children inherit a disease when both parents have the disease and share a similar genotype?

Select one:
- Mitochondrial
- Autosomal dominant
- X linked recessive
- X linked dominant
- Autosomal recessive

Your answer is incorrect.

If both parents are diseased and have an autosomal recessive condition, then all children will be diseased (100%). If both are diseased with heterozygous genotype, an autosomal dominant disease will be transmitted 3 out of 4 occasions to their children.

The correct answer is: Autosomal dominant

Question 50
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 018
Which of the following specific phobias has a strongly genetic aetiology?

Select one:
- Spider phobia
- Acrophobia
- Animal phobia
- Space phobia
- Blood injury injection phobia

Your answer is incorrect.

Blood injury injection phobia is oddly different from other phobias in that the response to exposure is not tachycardia and sympathetically driven heart rate, etc. Instead, a fainting response occurs where the patient may drop fainting with low BP and bradycardia.

The correct answer is: Blood injury injection phobia

Question 51
Not answered

MoK Neuroanatomy 001
Which of the following structures is involved in addictive behaviour?

Marked out of 1.00

Flag question

Select one:

- Suprachiasmatic nucleus
- Nucleus tractus solitarius
- Substantia nigra
- Ventral tegmentum
- Preoptic nucleus

Your answer is incorrect.

All forms of additive behaviors appear to be mediated through at least three structures: the ventral tegmental area VTA, the nucleus accumbens NA and the septal nucleus. VTA and NA contain neurons that use dopamine as their major neurotransmitter.

The correct answer is: Ventral tegmentum

Question 52

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 002
Which brain structure is spared during normal aging?

Select one:

- Cerebellum
- Motor neuron
- Cerebral cortex
- Sub thalamic nuclei
- Cranial nerves nuclei

Your answer is incorrect.

Cranial nerve nuclei are spared during the normal ageing process. All the other structures could be affected.

The correct answer is: Cranial nerves nuclei

Question 53

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 003
Loss of taste sensation in the anterior part of tongue is related to

Select one:

- 5th Nerve Palsy
- 9th Nerve Palsy
- 8th Nerve Palsy
- 6th Nerve Palsy
- 7th Nerve Palsy

Your answer is incorrect.

The facial nerve supplies the taste sensation in the anterior tongue. The chorda tympani (the branch mediating this sensation) leaves the parent nerve, crossing through the middle ear, where it can also be damaged by severe infections, etc.

The correct answer is: 7th Nerve Palsy

Question 54

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 004
The Kluver-Bucy syndrome is typically associated with lesions of which of the following structures?

Select one:

- Amygdala
- Medial dorsal thalamus
- Lateral hypothalamus
- Globus pallidus
- Subthalamic nucleus

Your answer is incorrect.

This syndrome is the result of lesions of the temporal lobe in which parts of the amygdala are involved. Lesions of other regions such as the hypo- thalamus, cingulate cortex, or septal area do not produce the Kluver-Bucy syndrome.

The correct answer is: Amygdala

Question 55

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 005
The last part of the brain to mature is

Select one:

- Posterior parietal pole
- Occipital pole
- Anterior parietal pole
- Anterior frontal pole
- Hippocampus

Your answer is incorrect.

The last area of the brain to develop is the frontal lobe. The last region of the brain to mature is pre-frontal cortex (anterior frontal pole). While it may be fully formed in size, various parts within the structure of the brain continue to develop well into a person's twenties. The adolescent brain is immature and in a dynamic state of development. Frontal brain development is also a factor related to impulsive behavior among teens.

The correct answer is: Anterior frontal pole

Question 56

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 001

Which of the following is the precursor of serotonin?

Select one:

- 5 hydroxy dopamine
- 5 Hydroxy indole acetic acid
- 5 Hydroxy tryptophan
- 5 Hydroxy tryptamine
- Anandamide

Your answer is incorrect.

Note that 5-hydroxy tryptamine is nothing but serotonin itself.

The correct answer is: 5 Hydroxy tryptophan

Question 57

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 002

The somatodendritic inhibition of 5-HT release is regulated by

Select one:

- 5-HT1A receptors
- 5-HT3 receptors
- 5-HT4 receptors
- 5-HT5 receptors
- 5-HT2 receptors

Your answer is incorrect.

The 5-HT1a receptor is found at both somatodendritic and post synaptic sites. Somatodendritic 5-HT1A receptors in the dorsal and median raphe nuclei are autoreceptors regulating 5-HT neuronal firing. Post synaptic 5HT1A receptors are found in the hippocampus and periaqueductal grey matter where they regulate behaviours such as resilience, impulsivity and restraint of excessive response to stress. The 5-HT1A receptor may be an important target in the action of antidepressants. (Ref: Fundamentals of clinical psychopharmacology- 18)

The correct answer is: 5-HT1A receptors

Question 58

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 003

A 70-year-old woman awakens with inability to walk and is found on the floor by her family. Her examination shows right face, arm, and leg weakness and numbness. A MRI shows no recent changes. What is the most likely cause of her gait problems?

Select one:

- Parkinsonism
- A right hemisphere infarction
- A right cerebellar infarction
- A left hemisphere infarction
- Drug Toxicity

Your answer is incorrect.

The presentation is suggestive of a cerebral infarct, most likely to be on the left side as the woman is presenting with right sided weakness

The correct answer is: A left hemisphere infarction

Question 59

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 004

Mr. P is a 77-year-old gentleman suffering from gradually progressive cognitive decline for last 6 months. He has history of Parkinson's disease for last 3 years and suffers from visual hallucinations. The chemical that is mainly involved and deficient in this condition is

Select one:

- Dopamine
- Acetyl choline
- Noradenaline
- GABA
- Serotonin

Your answer is incorrect.

The diagnosis is Parkinson's disease dementia. In both Lewy body dementia and Parkinson's disease dementia, brain acetyl choline levels are reduced similar to Alzheimer's disease. Therefore acetylcholinesterases (Anti-dementia drugs) could be used in the treatment of Lewy body dementia.

The correct answer is: Acetyl choline

Question 60
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 009
Hypothetical use of cholecystokinin has been studied in which of the following disorders?

Select one:
- Bulimia
- Conduct disorder
- Childhood schizophrenia
- Autism
- ADHD

Your answer is incorrect.
CCK can produce a sense of satiation bringing binges to a conclusion earlier than usual.
The correct answer is: Bulimia

Question 61
Not answered
Marked out of 1.00
Flag question

MoK Neuropathology 001
Hirano bodies seen in which one of the following types of dementia?

Select one:
- Vascular
- Lewy body type
- Pick's
- Frontotemporal
- Alzheimer's

Your answer is incorrect.
Hirano bodies were first described by Hirano in 1965. These are intracellular eosinophilic rod-shaped structures seen in the neurons of individuals with Alzheimer's and some forms of Creutzfeldt-Jacob disease (Cartier et al., 1985).
The correct answer is: Alzheimer's

Question 62
Not answered
Marked out of 1.00
Flag question

MoK Neuropathology 002
Tom is a 12-year-old boy with reduced social interactions, poor communication abilities and repetitive stereotypical behaviour. The macroscopic changes involved in the pathogenesis of this condition would be hypoplasia of the

Select one:
- Hippocampus
- Fusiform gyrus
- Cerebellar vermis
- Amygdala
- Angular gyrus

Your answer is incorrect.
Reduced social interactions, poor communication abilities and repetitive stereotypical behaviour are core features of autistic spectrum disorder. In children with autism, hypoplasia of cerebellar vermis and to some extent the cerebellar hemispheres is documented. Purkinje cell count in cerebellum is significantly lower
The correct answer is: Cerebellar vermis

Question 63
Not answered
Marked out of 1.00
Flag question

MoK Neuropathology 003
In CJD, the amyloid plaques are prominent in which part of the brain?

Select one:
- Cerebellum
- Raphe nuclei
- Corpus callosum
- Occipital cortex
- Hippocampus

Your answer is incorrect.
CJD is a type of multifocal dementia involving the cerebellar structures, where amyloid plaques are prominent.
The correct answer is: Cerebellum

Question 64
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 001
Normal adult enters deep sleep from wakefulness before entering REM sleep. But infants enter REM sleep directly from wakefulness. When does the adult pattern of sleep develop in a child?

Select one:

- 12 months
- 3 years
- 18 months
- 5 years
- 3 months

Your answer is incorrect.

Newborns sleep for 16 - 18 hours daily though the longest sleep episode lasts only 2.5 to 4 hours. Newborns have three types of sleep: quiet sleep (similar to NREM), active sleep (analogous to REM), and indeterminate sleep. Sleep onset occurs through REM, unlike adults where it occurs through NREM. In infants, each sleep episode is made of only one or two cycles (instead of 4 or more in adults). This infantile sleep architecture disappears when circadian rhythms are fully entrained. Circadian rhythm development in the first 1 month involves the emergence of the 24-hour core body temperature cycle; by 2, months progression of nocturnal sleeping is noted and in 3 months, melatonin and cortisol rhythms are established along with NREM in the sleep onset phase. http://www.ncbi.nlm.nih.gov/books/NBK19956/

The correct answer is: 3 months

Question 65

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 002

A 40 year old man presents with gradually developing darkening of the skin most visible on scars, skin folds and pressure points. Following a trivial driving accident, he notices extreme fatigue, muscle weakness and weight loss.The most commonly used definitive test for this condition is

Select one:

- Skin Biopsy
- ELISA serology
- Prolactin Test
- ACTH stimulation test
- Fasting glucose

Your answer is incorrect.

This patient has Addison's disease which, due to insidious onset, often goes unnoticed until something else brings the symptoms to notice. The ACTH stimulation test is the most commonly used test for diagnosing adrenal insufficiency. In this test, blood cortisol, urine cortisol, or both are measured before and after a synthetic form of ACTH is given by injection. The normal response after an ACTH injection is a rise in blood and urine cortisol levels. People with Addison's disease or long-standing secondary adrenal insufficiency have little or no increase in cortisol levels.

The correct answer is: ACTH stimulation test

Question 66

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 003

During relaxed periods of wakefulness, a normal individual's EEG will display which of the following rhythms?

Select one:

- Delta rhythm
- Gamma rhythm
- Alpha rhythm
- Beta rhythm
- Theta rhythm

Your answer is incorrect.

During states of quiet wakefulness or drowsiness, the EEG pattern becomes slower (8 to 13 Hz with an average amplitude of 50 V) than what is seen during an alert state. This constitutes the alpha rhythm.

The correct answer is: Alpha rhythm

Question 67

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI001

Adverse effects

For each of the following drugs, identify the most important side effect associated with its therapeutic use

Procyclidine: Choose...

Ziprasidone: Choose...

Olanzapine: Choose...

Your answer is incorrect.

Procyclidine, Benztropine, Orphenadrine and Biperiden are anticholinergic drugs that can cause dry mouth, constipation, urinary retention and blurred vision. The latter is associated with impaired visual accommodation, causing difficulty in reading.

There are concerns over the use of Ziprasidone, as it can increase QTc. However with this drug, EPSE is equivalent to placebo, no appreciable weight gain, and no appreciable anticholinergic side effects are noted.

Although possible glucose dysregulation and metabolic syndrome are reported with most atypical antipsychotics, Olanzapine is the most common offender. Metabolic syndrome is a cluster of disorders comprising of obesity(central and abdominal), dyslipidaemias, glucose intolerance, insulin resistance (or hyperinsulinaemia) and hypertension.

The correct answer is: Procyclidine – Blurred vision, Ziprasidone – QTc prolongation, Olanzapine – Glucose dysregulation

Question 68

Not answered

MoK Pharmacology EMI002

Mechanism of drug action

Find the drug that matches each of the following mechanisms of action;

Marked out of 3.00

Flag question

Phosphodiesterase-5 inhibitor Choose...

Butyrylcholinesterase inhibitor Choose...

Dopamine reuptake inhibitor Choose...

Your answer is incorrect.

Sildenafil is a potent and selective inhibitor of c-GMP- specific phosphodiesterase type 5 (PDE5), which is responsible for degradation of c-GMP in corpus cavernosum of the penis. Rivastigmine inhibits both acetyl and butyrylcholinesterases. Bupropion is a DARI - Dopamine reuptake inhibitor

The correct answer is: Phosphodiesterase-5 inhibitor – Sildenafil, Butyrylcholinesterase inhibitor – Rivastigmine, Dopamine reuptake inhibitor – Bupropion

Question 69

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI003

Swapping antidepressants

For each of the following change of antidepressants, choose ONE method of cross tapering from options. Each option may be used once, more than once or not at all.

Venlafaxine to mirtazapine in a patient who has developed severe gastrointestinal intolerance. Choose...

Fluoxetine to tranylcypromine due to the inefficacy of the former. Choose...

Paroxetine to fluoxetine due to anticholinergic side effects of the former Choose...

Your answer is incorrect.

All answers are based on Maudsley Prescribing Guidelines, 10th edition.

Venlafaxine has a short half-life. Incases where side effects are intolerable, it must be withdrawn promptly but the second agent mirtazapine should be started immediately. The risk of serotonin syndrome is not a significant problem with this approach with VFX and MZP being often used together as combination therapy. A similar approach can also be used when stopping one SSRI and starting another, as in case 3. When administered together MAOI and SSRI can increase the risk of the serotonin syndrome manifold.

Fluoxetine has a long half-life, and its metabolite stays in the system for a much longer time (2-4 weeks). To be safe, it is recommended that a five weeks waiting period is applied before MAOIs are started after a patient has discontinued fluoxetine.

The correct answer is: Venlafaxine to mirtazapine in a patient who has developed severe gastrointestinal intolerance. – Withdraw the first and start the second immediately after stopping it, Fluoxetine to tranylcypromine due to the inefficacy of the former. – Withdraw, wait for five weeks and then start the second drug, Paroxetine to fluoxetine due to anticholinergic side effects of the former – Withdraw the first and start the second immediately after stopping it

Question 70

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI001

Research studies in genetics

Select the most appropriate type of study for each description given below

An investigator is interested in calculating the risk of illness within families among first, second and third degree relatives and compare the rates with the established risk in the general population. Choose...

An investigator wants to compare the likelihood of the observed rate of illness in a given pedigree with those families that have a differing mode of transmission Choose...

A researcher has no candidate genes of a priori interest but wants to study the whole genome in large samples of patients and controls to investigate the most commonly observed polymorphic variations in the patients. Choose...

Your answer is incorrect.

Family studies aid in determining the likelihood of illness among relatives. **Segregation analysis** is used to investigate whether familial data for a disorder or trait suggests any particular modes of inheritance. Both monogenic (Mendelian) or polygenic models can be tested in this fashion. In addition, age of onset and sex effects can be taken into account besides the specific genetic hypothesis under consideration. In **Genome-Wide Association Studies** (GWAS), a 'candidate gene' approach is not used. Instead, several thousands of single nucleotide polymorphisms are assayed in thousands of individuals.

The correct answer is: An investigator is interested in calculating the risk of illness within families among first, second and third degree relatives and compare the rates with the established risk in the general population. – Family studies, An investigator wants to compare the likelihood of the observed rate of illness in a given pedigree with those families that have a differing mode of transmission – Segregation analysis, A researcher has no candidate genes of a priori interest but wants to study the whole genome in large samples of patients and controls to investigate the most commonly observed polymorphic variations in the patients. – Genome-wide association studies

Question 71

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI002

Bedside tests of cognition

For each of the following options below, choose the most appropriate test from the given list:

A test of visual memory where a child is shown ten geometric designs, one at a time, and asked to reproduce each one as exactly as possible on plain paper from memory. Choose...

A test to check inhibitory control aspect of executive function Choose...

This test measures the recall of learned material. It is a complex figure test, which distinguishes between brain damage and mental retardation. Choose...

Your answer is incorrect.

Rey-Osterrieth complex figure test is one of the non verbal memory tests. Here the subject is first asked to copy a complex geometric figure and then to draw from memory after an interval of 30 minutes. The recall is impaired in patients with dementia and amnesic syndrome.

The Benton Visual Retention Test (aka Benton Test) is used for individuals aged 8 onwards to measure visual perception and visual memory. It can also help evaluating possible learning disabilities. The subject is asked to draw from memory each one of the ten designs that are shown one at a time.

When the name of a colour (e.g., "blue," "green," or "red") is written in a colour that is different from the name (e.g., the word "red" printed in blue ink instead of red ink), naming the actual colour of the colour-incongruent word is challenging, takes longer and is more error-prone than namign the colour of a congruent word. This is called Stroop Effect and is utilised in testing the executive faculty of inhibition (suppression of interferences) and error correction.

The correct answer is: A test of visual memory where a child is shown ten geometric designs, one at a time, and asked to reproduce each one as exactly as possible on plain paper from memory. – Benton test, A test to check inhibitory control aspect of executive function – Stroop test, This test measures the recall of learned material. It is a complex

figure test, which distinguishes between brain damage and mental retardation. – Rey-Osterreith test

Question 72
Not answered
Marked out of 6.00
Flag question

MoK Neurosciences CXEMI003

Localizing signs
Choose two symptoms for each of the following conditions:

1. A malnourished alcoholic. (TWO) [ ] [ ]

2. A right-handed person with right parietal lobe lesion. (TWO) [ ] [ ]

3. A frontal meningioma. (TWO) [ ] [ ]

Explanation

Case 1: This patient is most likely to have Wernicke's encephalopathy. The classic triad includes ophthalmoplegia, ataxia and confusion. The most common ophthalmoplegia here is a bilateral 6th nerve (not 4th nerve) lesion. The 4th nerve may be affected in few cases.

Case 2: The non-dominant parietal lobe has a significant role in constructional praxis and also in awareness of illness states.

Case 3: Frontal meningiomas are associated with anosmia (often present due to the common olfactory groove distribution of tumours causing pressure) and ipsilateral optic atrophy, on the side of the tumour. Anosmia with ipsilateral optic atrophy and contralateral papilledema is also known as Kennedy-Foster syndrome.

The correct answer is:
A malnourished alcoholic. (TWO) - Gait Ataxia , Bilateral 6th nerve palsy,

A right-handed person with right parietal lobe lesion. (TWO) - Anosognosia, Constructional Apraxia,

A frontal meningioma. (TWO) - Ipsilateral optic atrophy , Anosmia

Question 73
Not answered
Marked out of 5.00
Flag question

MoK Neurosciences EMI004
Inheritance of psychosis
A 26-year-old woman has been admitted to hospital for bizarre delusions, auditory hallucinations and disorganized speech. She is also flat in her affect, unkempt and is self-neglecting. There is no history of mania or depression. For each question below, choose the correct answer from the given list:

What is the chance of her 12-year-old son inheriting the disorder? Choose...

She has a sister who has a 6-year-old son with Down's syndrome. What is the chance of him developing the same psychotic disorder? Choose...

What is the chance of her identical twin sister to inherit the disorder? Choose...

If her child's father is also suffering from the same psychotic disorder, what is the risk of inheritance in the child? Choose...

What is the risk of her non-identical twin brother to inherit the disorder? Choose...

Your answer is incorrect.

Siblings and offspring of patients with schizophrenia have about a 10-fold increase in their lifetime risk of developing the disorder compared to the general population, whereas in second- degree relatives (e.g. nephews, nieces, grandchildren) the risk is about 3-4 times the population risk. Children with one parent affected by schizophrenia have a 13% risk of developing the illness. The risk increases with the number of relatives affected, rising from 9% in a sibling to 16% if a parent and a sibling are affected and 45% if both parents are affected. A meta-analysis of a large number of European twin studies has produced a monozygotic concordance rate of 46% compared with a dizygotic concordance rate of 14%. Patients with Down's syndrome have lower that the average population risk for schizophrenia (choose the lowest number - option A).

The correct answer is: What is the chance of her 12-year-old son inheriting the disorder? – 11-15%, She has a sister who has a 6-year-old son with Down's syndrome. What is the chance of him developing the same psychotic disorder? – 1-5%, What is the chance of her identical twin sister to inherit the disorder? – 41-45%, If her child's father is also suffering from the same psychotic disorder, what is the risk of inheritance in the child? – 41-45%, What is the risk of her non-identical twin brother to inherit the disorder? – 11-15%

Question 74
Not answered
Marked out of 3.00
Flag question

MoK Neurosciences EMI005
Chromosomal profiles
Identify the karyotypic / genetic profile for each of the following conditions:

Mr. X is a 21-year-old tall, thin man with a moderate level of learning disabilities especially with reading. He has diabetes and presents with behavioural outbursts. Choose...

Miss. Y is a 19-year-old woman with a moderate learning disability and has a developed a schizophrenia-like psychosis. She has cardiac problems and cleft palate Choose...

Mr. Z is an 11-year-old boy with growth retardation who is very sociable with markedly better verbal than spatial intelligence. He has cardiac problems and sensory-motor impairments Choose...

Your answer is incorrect.

Case 1 describes a man with Klinefelter syndrome. Klinefelter syndrome (KS) (47, XXY) is the most common allosomal disorder in man and is a relatively frequent cause of male infertility and hypogonadism. Men with 47XXY makeup have an increased risk of diabetes and metabolic syndrome related to adiponectin dysregulation.

Case 2 is descriptive of Velocardio Facial Syndrome (VCFS). VCFS was first described by Kirkpatrick and DiGeorge in 1968 as a constellation of immunologic deficiencies (hence the name DiGeorge syndrome). It was renamed Shprintzen Syndrome following a thorough description of the common presenting symptoms by Shprintzen, et al., in 1978 who included palate anomalies (Velo), congenital cardiovascular defects (Cardio), and mild facial dysmorphism (Facial) as characteristic features of VCFS . Nearly 25% develop psychosis. It is associated with a microdeletion in the long arm of chromosome 22 at band 22q11.

Williams's syndrome (case 3) is a developmental disorder classically characterized by the triad of typical facial features, infantile hypercalcemia, and supra valvular aortic stenosis. Deletion of elastin gene at chromosome 7q11 is associated with this syndrome.
Ref: Morbidity and mortality in Klinefelter syndrome (47,XXY), http://www.ncbi.nlm.nih.gov/pubmed/21414026 (accessed April 5, 2015).

The correct answer is: Mr. X is a 21-year-old tall, thin man with a moderate level of learning disabilities especially with reading. He has diabetes and presents with behavioural outbursts. – 47 XXY, Miss. Y is a 19-year-old woman with a moderate learning disability and has a developed a schizophrenia-like psychosis. She has cardiac problems and cleft palate – Deletion of part of chromosome 22, Mr. Z is an 11-year-old boy with growth retardation who is very sociable with markedly better verbal than spatial intelligence. He has cardiac problems and sensory-motor impairments – Deletion of part of chromosome 7

Question 75

Not answered

Marked out of 5.00

Flag question

MoK Neurosciences EMI014

Neurotransmitters & disorders

Identify the disorders associated with each of the neurotransmitters listed below

Substance P	Choose...
Neurotensin	Choose...
Neuropeptide Y	Choose...
Corticotropin-releasing factor	Choose...
Endocannabinoids	Choose...

Your answer is incorrect.

Substance P is the primary neurotransmitter in most primary afferent sensory neurons (Pain) and is also seen in the striatonigral pathway. Levels of substance P have been reported to be markedly reduced in patients with Huntington's chorea.

Neurotensin coexists with dopamine in some axon terminals. Neurotensin-related peptides are being investigated in the treatment of psychotic symptoms.

Neuropeptide Y may play an important role in feeding disorders through its effects on energy balance. It is secreted by the hypothalamus and in addition to increasing food intake, it increases the proportion of energy stored as fat and blocks nociceptive signals to the brain.

Corticotropin releasing factor (CRF) is the neurotransmitter associated with anxiety and depression, and CRF receptor antagonists are currently under clinical evaluation as anxiolytics and antidepressants.

The endocannabinoids are associated with schizophrenia and psychotic disorders. Anandamide (weak ligand) and 2-arachnidonylglycerol (strong) are formed from arachidonic acid and ethanolamine - belong to endocannabinoid family. The two types of cannabinoid receptors, central (CB1) and peripheral (CB2), bind tetrahydrocannabinol (THC), the active ingredient of marijuana. CB1 receptors are found in the brain and are potential targets for the treatment of both pain and psychosis. CB2 receptors are associated with immune regulation.

The correct answer is: Substance P – Huntington's disease, Neurotensin – Schizophrenia, Neuropeptide Y – Feeding, Corticotropin-releasing factor – Depression, Endocannabinoids – Schizophrenia

Finish review